# Syllabus 9758 Statistics Example Questions

# Permutations and Combinations

## Choosing with "at least" conditions; circular arrangements

**Example 1** [TJC 2025 Prelim P2 Q6]

A treasure hunt game is to be played in a 3-storey building. There are 12 specific locations where 5 treasures are to be hidden. Among the 5 treasures, 4 of them are identical. The locations are distributed across the storeys as follows:

- 3 locations on the 1st storey,
- 3 locations on the 2nd storey, and
- 6 locations on the 3rd storey.

Each location can hold at most one treasure.

(a) Find the number of ways to distribute the 5 treasures among the 12 locations for each of the following separate cases:

(i) The 5 treasures can be hidden in any of the 12 locations. [1]

(ii) Exactly 2 treasures must be hidden on the 3rd storey. [2]

(iii) There must be at least one treasure on each of the 1st and 2nd storeys, and at least 2 treasures on the 3rd storey. [3]

Five teams of 2 players each take part in the game. The 10 players and 2 game masters are to be seated around a round table for a briefing.

(b) Find the number of different seating arrangements such that players from the same team must be seated next to each other but the two game masters must not be seated next to each other. [3]

**Official solution**

(a)(i) Required number of ways $= \binom{12}{5}\binom{5}{1} = 3960$

(a)(ii) Required number of ways $= \binom{6}{2}\binom{6}{3}\binom{5}{1} = 1500$

**Alternative method:**

Case 1: 2 not identical at 3rd storey $\binom{6}{2} \times 2! \binom{6}{3} = 600$

Case 2: 2 identical at 3rd storey $\binom{6}{2}\binom{6}{3}\frac{3!}{2!}=900$

(a)(iii) Case 1: 2 in 1st storey, 1 in 2nd storey, 2 in 3rd storey

Number of ways $=\binom{3}{2}\binom{3}{1}\binom{6}{2}\binom{5}{1}=675$

Case 2: 1 in 1st storey, 2 in 2nd storey, 2 in 3rd storey

Number of ways $=\binom{3}{1}\binom{3}{2}\binom{6}{2}\binom{5}{1}=675$

Case 3: 1 in 1st storey, 1 in 2nd storey, 3 in 3rd storey

Number of ways $=\binom{3}{1}\binom{3}{1}\binom{6}{3}\binom{5}{1}=900$

Total number of ways $=2250$

(b) Number of ways to arrange the 10 participants $=(5-1)!(2!)^5=768$

Number of ways to slot in the 2 game masters $=\binom{5}{2}2!=20$

Required number $=768\times 20=15360$

**Example 2** [ASRJC 2025 Prelim P2 Q11(b)(iii)]

To run an event, the librarian plans to select a team of 12 volunteers from a group of 18. The volunteers comprise three age groups: 5 youths, 6 young adults, and 7 seniors.

(iii) In how many ways can the team be formed if at least one volunteer must be chosen from each age group? [3]

**Official solution**

(iii) Total number of ways $={}^{18}C_{12}=18564$

No youth $={}^{13}C_{12}=13$

No young adult $={}^{12}C_{12}=1$

Total possible ways $=18564-13-1=18550$

## Choosing teams (football)

**Example 3** [JPJC 2026 Prelim P2 Q8]

A group of 12 players is available for a street football match. The group consists of 5 attackers, 4 defenders and 3 goalkeepers. A team of 5 players is selected at random from the group.

(a) Find the

(i) number of different teams containing exactly 2 attackers, 2 defenders and 1 goalkeeper, [2]

(ii) probability that the team contains exactly 1 goalkeeper and at most 2 attackers. [3]

The coach has already decided which goalkeeper will play in a match. To help him select from the 5 attackers and 4 defenders, he conducts a passing drill involving these players. For each pass, 1 player is chosen at random to receive the ball. After each pass, the ball is returned to the coach before the next player is chosen. A player may receive the ball more than once. The coach makes 4 passes in this manner. Find the probability that

(b) exactly 3 passes are made to attackers, [2]

(c) either the first 3 passes or the last 3 passes are made to attackers (or both). [3]

**Official solution**

**(a)(i)** Number of teams $= {}^5C_2 \times {}^4C_2 \times {}^3C_1 = 180$

**(a)(ii)**

Case 1: 1 goalkeeper, 2 attackers, 2 defenders $\quad {}^3C_1 \times {}^5C_2 \times {}^4C_2 = 180$

Case 2: 1 goalkeeper, 1 attacker, 3 defenders $\quad {}^3C_1 \times {}^5C_1 \times {}^4C_3 = 60$

Case 3: 1 goalkeeper, 0 attacker, 4 defenders $\quad {}^3C_1 \times {}^5C_0 \times {}^4C_4 = 3$

Total number of ways $= {}^{12}C_5 = 792$

Hence probability that team has at most 2 attackers $= \dfrac{180+60+3}{792} = \dfrac{243}{792}$ or 0.307

**(b)** Req probability

$= \mathrm{P}(\mathrm{AAAD}) + \mathrm{P}(\mathrm{AADA}) + \mathrm{P}(\mathrm{ADAA}) + \mathrm{P}(\mathrm{DAAA})$

$= \left(\dfrac{5}{9}\right)^3 \left(\dfrac{4}{9}\right) \times 4 = \dfrac{2000}{6561}$ or 0.305

**(c)** Method 1:

Req probability $= \mathrm{P}(\mathrm{AAA_}) + \mathrm{P}(\mathrm{_AAA}) - \mathrm{P}(\mathrm{AAAA})$

$= \left(\dfrac{5}{9}\right)^3 + \left(\dfrac{5}{9}\right)^3 - \left(\dfrac{5}{9}\right)^4$

$\dfrac{1625}{6561}$ or 0.248

Method 2:

Req prob $= \mathrm{P}(\mathrm{AAAD}) + \mathrm{P}(\mathrm{DAAA}) + \mathrm{P}(\mathrm{AAAA})$

$= \left(\dfrac{5}{9}\right)^3 \left(\dfrac{4}{9}\right) + \left(\dfrac{4}{9}\right) \left(\dfrac{5}{9}\right)^3 + \left(\dfrac{5}{9}\right)^4$

$\dfrac{1625}{6561}$ or 0.248

## Arranging in rows ($2\times n$); round table

**Example 4** [ASRJC 2024 Prelim P2 Q8]

An office team of 10 people includes 7 men and 3 women named Anne, Beth, and Cathie. For an upcoming fire drill exercise, 5 individuals will be chosen, each assigned a unique role, to carry out the drill. Determine the number of possible ways to select 5 people from this group of 10

(i) to conduct the fire drill, [1]

(ii) such that at most 1 woman is selected to conduct the fire drill. [2]

(iii) After the fire drill exercise, the 10 people are to hold a discussion at a round table with 10 identical seats. Determine the number of ways in which Beth is seated between Anne and Cathie. [1]

(iv) A group photo of the 10 people, arranged in two rows of five, was taken after the discussion. Determine the number of ways in which Beth is not standing beside Anne or Cathie. [4]

**Official solution**

**(i)** Number of ways $= \binom{10}{5}\times 5! = 30240$

**(ii)** Number of ways $= \left[\binom{7}{5}\binom{3}{0}+\binom{7}{4}\binom{3}{1}\right]5!$

$= (21\times 1+35\times 3)5! = 15120$

**(iii)** Number of ways $= (8-1)!2! = 10080$

**(iv)** Case 1 – Beth in 1 row while Anne and Cathie are in another row

$\binom{7}{4}\times 5!\times 5!\times 2 = 1008000$

Case 2 – Beth and one of them in 1 row

$\binom{7}{3}\times 3!\times \binom{2}{1}\times \binom{4}{2}\times 2!\times 5!\times 2 = 1209600$

Case 3 – Anne, Beth and Cathie are in the same row

$\underbrace{\binom{7}{2}\times 2!\times \binom{3}{2}\times 2!\times 2!\times 5!\times 2}_{\text{A and C together}} + \underbrace{\binom{7}{2}\times 2!\times 3!\times 5!\times 2}_{\text{A and C separated}} = 181440$

Thus number of ways $= 1008000+1209600+181440 = 2399040$

# Probability

## Tree diagram

**Example 5** [EJC 2025 Prelim P2 Q6]

An online game consists of three stages. All players start with Stage 1, and no matter whether they win or lose, they proceed to Stage 2. Players are allowed to proceed to Stage 3 only if they have won at least one of the first two stages.

Ming's probability of winning Stage 1 is $p$.

If Ming wins a stage, his probability of winning the next stage is halved. If he loses a stage, his probability of winning remains unchanged in the next stage.

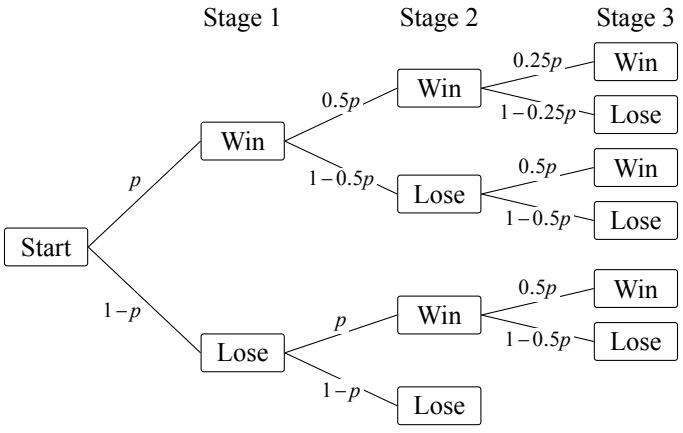
(a) Draw a probability tree diagram to represent all possible outcomes for Ming over the three stages. [3]

(b) Find the range of possible values of $p$ if the probability that Ming wins all three stages is more than 0.1. [2]

(c) It is now known that $p = 0.9$. Given that Ming proceeds to Stage 3, find the probability that he wins exactly two stages out of the three. [4]

**Official solution**

**(a)**




**(b)** $p(0.5p)\ (0.5^2p) > 0.1$

$p^3 > 0.8$

$p > 0.928$

**(c)** $\mathrm{P}(\text{wins exactly 2 stages out of 3} \mid \text{proceeds to Stage 3})$

$$= \frac{\mathrm{P}(\text{wins exactly 2 stages out of 3} \cap \text{proceeds to Stage 3})}{\mathrm{P}(\text{proceeds to Stage 3})}$$

$$= \frac{0.9(0.45)(1-0.225)+0.9(0.55)(0.45)+0.1(0.9)(0.45)}{1-(0.1)(0.1)}$$

$$= \frac{0.577125}{0.99} = 0.583$$

## Venn diagram: range of values of a probability

**Example 6** [CJC 2024 Prelim P2 Q10(b)]

The probability of events $A$, $B$ and $C$ occurring are given by $\mathrm{P}(A) = \frac{8}{25}$, $\mathrm{P}(B) = \frac{61}{100}$ and $\mathrm{P}(C) = \frac{7}{20}$ respectively. It is also known that $\mathrm{P}(A \cap C) = \frac{11}{100}$, $\mathrm{P}(B \mid C) = \frac{13}{35}$ and $\mathrm{P}(A \cap B \cap C) = \frac{7}{100}$.

(i) Find $\mathrm{P}(B \cap C)$. [2]

(ii) If events $A$ and $B$ are independent, find $\mathrm{P}(A \cap B \cap C')$. [2]

(iii) If events $A$ and $B$ are **not** known to be independent, find the greatest and least possible values of $\mathrm{P}(A \cap B \cap C')$. [4]

**Official solution**

(bi) $\mathrm{P}(B \mid C) = \frac{13}{35}$

$$\frac{\mathrm{P}(B \cap C)}{\mathrm{P}(C)} = \frac{13}{35}$$

$$\frac{\mathrm{P}(B \cap C)}{\frac{7}{20}} = \frac{13}{35}$$

$$\mathrm{P}(B \cap C) = \frac{13}{100}$$

(bii) $\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$

$$= \frac{8}{25} \times \frac{61}{100}$$

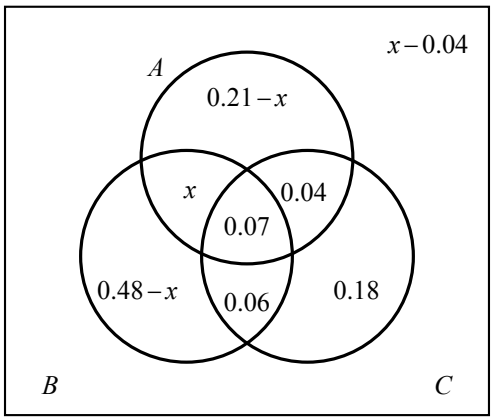
$$= \frac{122}{625}$$

$$\mathrm{P}(A \cap B \cap C')$$

$$= \frac{122}{625} - \frac{7}{100}$$

$$= \frac{313}{2500}$$

(biii)




Let $\mathrm{P}(A\cap B\cap C')=x$ where $0\leqslant x\leqslant 1$ … (1)

Then

$$\mathrm{P}(A\cap B'\cap C')=\frac{8}{25}-\frac{11}{100}-x=\frac{21}{100}-x$$

$$\mathrm{P}(A'\cap B\cap C')=\frac{61}{100}-\frac{13}{100}-x=\frac{12}{25}-x$$

$$\mathrm{P}(A\cup B\cup C)$$

$$=\mathrm{P}(C)+\mathrm{P}(A\cap B'\cap C')+\mathrm{P}(A'\cap B\cap C')+\mathrm{P}(A\cap B\cap C')$$

$$=\frac{7}{20}+\left(\frac{21}{100}-x\right)+\left(\frac{12}{25}-x\right)+x$$

$$=\frac{26}{25}-x$$

Consider the probabilities $\mathrm{P}(A\cap B'\cap C')$, $\mathrm{P}(A'\cap B\cap C')$ and $\mathrm{P}(A\cup B\cup C)$,

$$0\leqslant \frac{21}{100}-x\leqslant 1 \quad \text{and} \quad 0\leqslant \frac{12}{25}-x\leqslant 1 \quad \text{and} \quad 0\leqslant \frac{26}{25}-x\leqslant 1$$

$$-\frac{79}{100}\leqslant x\leqslant \frac{21}{100} \text{ … (2)} \qquad -\frac{13}{25}\leqslant x\leqslant \frac{12}{25} \text{ … (3)} \qquad \frac{1}{25}\leqslant x\leqslant \frac{26}{25} \text{ … (4)}$$

Taking intersection of inequalities (1) to (4), $\frac{1}{25}\leqslant x\leqslant \frac{21}{100}$

$\min \mathrm{P}(A\cap B\cap C')=\frac{1}{25}$ and $\max \mathrm{P}(A\cap B\cap C')=\frac{21}{100}$

## Sum to infinity of a GP

**Example 7** [ACJC 2023 Prelim P2 Q9(a)]

A bag contains one green disc, $r$ red discs and $2r$ blue discs, where $r>1$.

(a) Tim and Sim play a game by taking turns to draw a disc from the bag at random. The drawn disc is replaced after each turn. The first person to draw the green disc wins. If Sim started the game first, show that the probability of Sim winning the game is $\frac{3r+1}{6r+1}$. [2]

**Official solution**

9(a) $\dfrac{1}{3r+1}+\left(\dfrac{3r}{3r+1}\right)^2\times\dfrac{1}{3r+1}+\left(\dfrac{3r}{3r+1}\right)^4\times\dfrac{1}{3r+1}+\ldots$

$=\dfrac{\frac{1}{3r+1}}{1-\left(\frac{3r}{3r+1}\right)^2}=\dfrac{3r+1}{6r+1}$

## Contextual probability

**Example 8** [HCI 2025 Prelim P2 Q6]

A basketball free throw game involves a team of two players, Ben and John. The game consists of at most three throws and the moment a player makes a successful shot, the team wins, and the game will end.

The game uses the following rules.

- Only one player is selected for each game and the probability that Ben is selected in a game is 0.7.
- The probabilities that Ben and John make a successful shot in any single attempt are 0.1 and 0.07 respectively.
- The shots are independent of each other.

(a) Find the probability that the team wins the game. [3]

(b) The team did not win the game. Find the probability that John was the one who was selected to shoot. [3]

(c) The team attempts the game repeatedly until the first game is won. Find the least number of attempts required such that the probability of winning within $n$ games is at least 0.95. [2]

**Official solution**

**(a)** $\mathrm{P}(\text{wins})$

$=\mathrm{P}(\text{wins and Ben selected})+\mathrm{P}(\text{wins and John selected})$

$=\mathrm{P}(\text{wins}\,|\,\text{Ben selected})\mathrm{P}(\text{Ben selected})$

$+\mathrm{P}(\text{wins}\,|\,\text{John selected})\mathrm{P}(\text{John selected})$

$=(0.1+0.9\times0.1+0.9^2\times0.1)\,(0.7)$

$+(0.07+0.93\times0.07+0.93^2\times0.07)\,(0.3)$

$=0.2483929$

OR

$$\begin{aligned}
&\mathrm{P}(\text{wins}) \\
&= \mathrm{P}(\text{wins and Ben selected}) + \mathrm{P}(\text{wins and John selected}) \\
&= \mathrm{P}(\text{wins} \mid \text{Ben selected})\mathrm{P}(\text{Ben selected}) + \\
&\quad \mathrm{P}(\text{wins} \mid \text{John selected})\mathrm{P}(\text{John selected}) \\
&= (1-0.9^3)\,(0.7) + (1-0.93^3)\,(0.3) \\
&= 0.2483929
\end{aligned}$$

**(b)**
$$\begin{aligned}
&\mathrm{P}(\text{John selected} \mid \text{did not win}) \\
&= \frac{\mathrm{P}(\text{John selected and did not win})}{\mathrm{P}(\text{did not win})} \\
&= \frac{0.3 \times 0.93^3}{1-0.2483929} \\
&= 0.321 \text{ (3sf)}
\end{aligned}$$

**(c)** Required probability $= 1 -$ P(did not win in all $n$ games)

$$= 1 - (1-0.2483929)^n$$

OR

Required probability

$$\begin{aligned}
&= \mathrm{P}(\text{win in 1st game}) + \mathrm{P}(\text{win in 2nd game}) \\
&\quad + \mathrm{P}(\text{win in 3rd game}) + \ldots + \mathrm{P}(\text{win in } n^{\text{th}} \text{ game}) \\
&= 0.2483929 + (1-0.2483929)(0.2483929) \\
&\quad + (1-0.2483929)^2(0.2483929) + \ldots \\
&\quad + (1-0.2483929)^{n-1}(0.2483929) \\
&= \frac{0.2483929\,(1-(1-0.2483929)^n)}{1-(1-0.2483929)} \\
&= 1-(1-0.2483929)^n
\end{aligned}$$

$$1-(1-0.2483929)^n \geqslant 0.95$$

$$(1-0.2483929)^n \leqslant 0.05$$

| $n$ | $(1-0.2483929)^n$ |
|---|---|
| 10 | $0.0575 > 0.05$ |
| 11 | $0.0432 < 0.05$ |

Smallest $n$ is 11.

*See also Example 3 (b), (c) for another contextual probability question.*

# Discrete Random Variables

## Expectation of a new variable

**Example 9** [SAJC]

The number of muffins, $X$, bought by a randomly chosen customer at a muffin shop has probability distribution as shown in the table below:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | $>6$ |
|---|---|---|---|---|---|---|---|
| $\mathrm{P}(X=x)$ | $\frac{1}{12}$ | $\frac{2}{12}$ | $\frac{3}{12}$ | $a$ | $\frac{2}{12}$ | $\frac{1}{12}$ | 0 |

(i) Find the value of $a$.

(ii) State the expected value of $X$.

(iii) Evaluate the variance of $X$.

Initially, the shopkeeper charges \$2.50 for each muffin, which cost 50 cents to produce.

(iv) Using the answer to part (ii), or otherwise, find the expected profit made per customer.

The shopkeeper decides to promote the sale of muffins by giving a free muffin to all customers who buy 4 or more muffins. As a result, the probability distribution of the number of muffins, $Y$, taken away by a customer is given in the following table.

| $y$ | 1 | 2 | 3 | 4 | 5 | 6 | $>6$ |
|---|---|---|---|---|---|---|---|
| $\mathrm{P}(Y=y)$ | $\frac{3}{36}$ | $\frac{4}{36}$ | $\frac{3}{36}$ | 0 | $\frac{20}{36}$ | $\frac{6}{36}$ | 0 |

(v) Find the expected profit made per customer.

(vi) If the number of customers increases by 8%, is the promotion more profitable for the shopkeeper? Give a reason for your answer.

[Ans: (i) $\frac{1}{4}$ (ii) 3.5 (iii) $1\frac{11}{12}$ (v) \$6.86 ]